QML – RAG: A Quantum-Classical Continual Retrieval Augmented Generation for Clinical Decision Support

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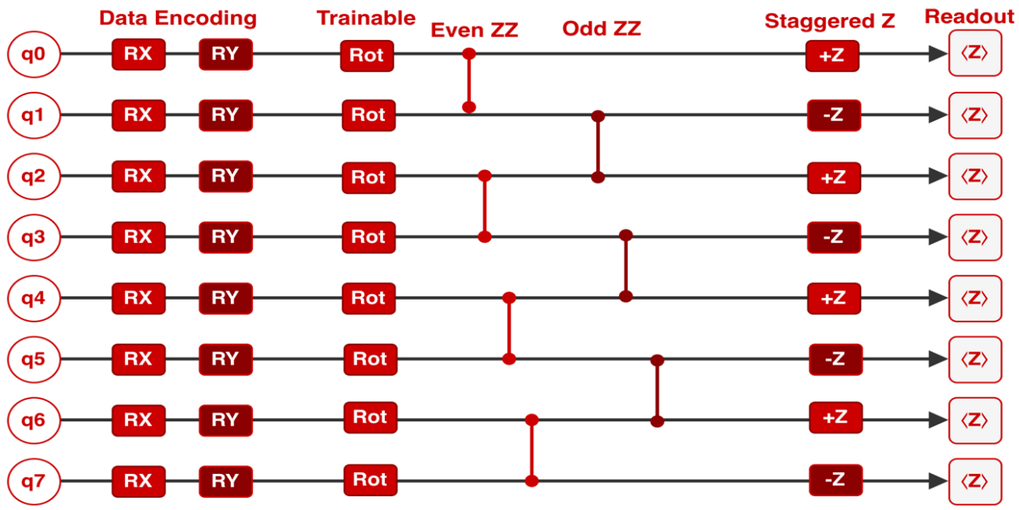
We combine Machine Learning (ML) in a virtual quantum computing (VQC) environment with a Retrieval Augmented Generation (RAG) pipeline leveraging large language models (LLMs) to generate medical treatment guidelines for patients in an intensive care unit (ICU). Medical chart data from high-priority patients are presented to a RAG pipeline that leverages the PMC-Patients database to match the record with relevant medical research from the PubMed database in order to generate a targeted treatment plan based on differential diagnosis and research-based interventions. Patients flagged by the QML as high-risk are processed through an iterative, embedding-driven RAG pipeline. Treatment recommendations are passed to a separate LLM to simulate expected physiological responses. Updated patient responses are re-evaluated with QML to produce a new risk classification reflecting the modeled impact of the intervention.

Quantum ML

Our pipeline uses a binary categorization ML model to prioritize patients from a set of anonymized ICU records from the MIMIC collection. We train an 8-qubit symmetry-aware QNN using IBM Qiskit with IsingZZ open-chain entanglement. 20 clinical features from MIMIC-III are compressed via PCA into 8 principal components, scaled to [0, π], and encoded as qubit rotation angles. All 8 qubits are measured via staggered Pauli-Z readout, breaking bitwise negation symmetry.

Data Source:

MIMIC-III Clinical Database: 135 ICU patients · 48-hour window · 20 high-risk patients



8-qubit IsingZZ variational circuit — checkerboard entanglement and staggered Z readout break bitwise negation symmetry

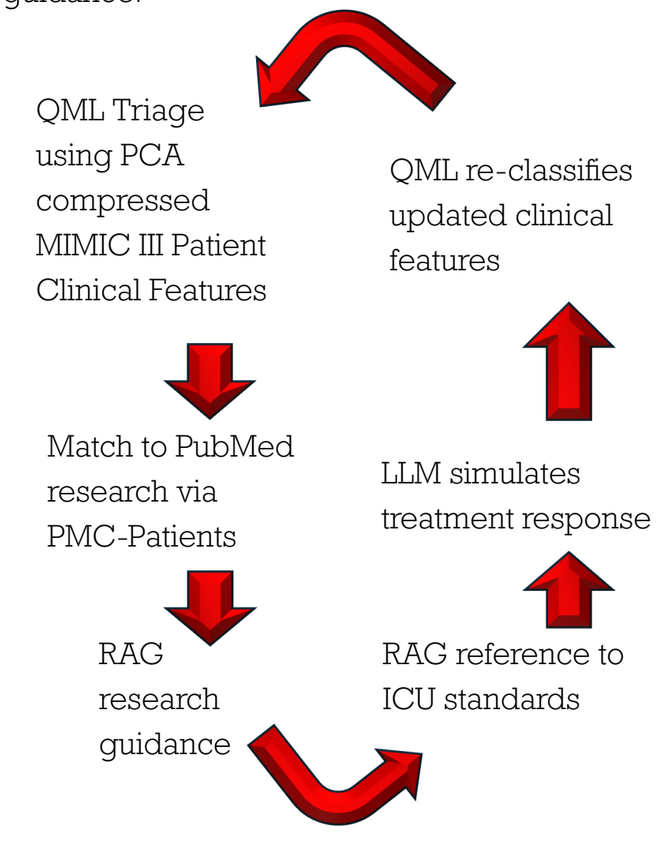
20 Clinical Features:

Vitals: HR · SpO_2 · MAP · SBP · DBP · RR · Temp
Labs: Glucose · WBC · Hematocrit · Platelets · Potassium · Sodium · Creatinine · BUN · Bicarbonate · Lactate
Derived: PaO_2/FiO_2 · GCS · Urine Output

Johnson, A., Pollard, T., & Mark, R. (2016). MIMIC-III Clinical Database (version 1.4). PhysioNet. RRID:SCR_007345. https://doi.org/10.13026/C2XW26

Augmented Knowledge

High-risk patients flagged by the QML proceed to our RAG pipeline. Patient snapshots — enriched with MIMIC-III diagnoses and chart data — are embedded and matched against research literature in PMC-Patients. Relevant PubMed abstracts generate evidence-based treatment guidance.



Research prototype · Not a medical device
No autonomous diagnosis or treatment
Clinician verification required